Should States Encourage Federal Benefit Takeup?

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Abstract

States often control the process for claiming benefits that are largely financed by the federal government. I study how this division of responsibility changes the marginal value of public funds (MVPF) from the perspective of the state making the participation decision. I separate the return to the state and local treasury from the value to state residents, who still bear some of the federal cost. The calculations begin with existing welfare accounts. I reallocate their fiscal components across governments while holding fixed, where possible, the causal estimates and the authors' treatment of willingness to pay. This substantially changes several of the results. In a Pennsylvania SNAP experiment, two consolidated MVPFs below one become state-resident values close to ten. California's move to semiannual SNAP reporting pays for itself for residents under the included components if enough of the assumed administrative saving could actually be realized. For Virginia's CommonHelp portal, a reconstruction calibrated to the authors' figure produces a state-resident benchmark of 4.60, compared with their consolidated estimate of 1.27. These calculations change the fiscal denominator. They leave open the separate question of how marginal participants value enrollment.

1. Introduction

Much of the practical design of a transfer program is left to the states. A state chooses the application form, how often a recipient must recertify, whether to send a reminder letter, and how its online portal works. In some settings it also chooses the default that follows an eligibility determination. These choices influence whether eligible people receive benefits. At the same time, the federal government may pay most of the resulting benefits, while the state or a county bears the cost of changing the application process. This division of responsibilities raises a simple question: how should a state evaluate a policy that makes a federally financed benefit easier to claim?

The usual MVPF calculation combines the fiscal effects on all levels of government. That is the appropriate account for many purposes, and I continue to report it as the national benchmark. A state deciding whether to change its own administrative policy faces a different fiscal account. Its state and local governments bear the costs recorded in those accounts, receive any savings there, and represent residents who bear only a share of the federal cost. This paper works out that narrower account and applies it to several policies intended to increase or maintain benefit participation.

The Pennsylvania SNAP experiment studied by Finkelstein and Notowidigdo (2019) previews the size of the difference. Sending information to likely eligible older adults has a consolidated MVPF of about 0.89. Adding application assistance produces an estimate of about 0.91 after including the resources used to provide that assistance. I obtain corresponding state-resident values of 10.57 and 9.78 when Pennsylvania's own costs are counted in full and federal costs are allocated to its residents using the state's population share. If the boundary is narrowed again to the state and local treasury, the values are 19.25 and 16.35.

These three calculations have different interpretations. The national calculation counts the full federal outlay wherever the recipients happen to live. The treasury calculation compares the welfare accruing to Pennsylvania residents with the dollars recorded in Pennsylvania state and local accounts. Federal benefit payments lie outside this budget boundary. The state-resident calculation adds back the portion of federal spending ultimately borne by Pennsylvania residents. Thus, federal financing is inexpensive from the perspective of a small state, although some of its cost still returns to the state's residents through the federal fiscal system.

Take-up policies make this separation especially important. A state can mail a letter that induces a federal SNAP payment. A county may remove a recertification step, save some worker time, and extend federally financed benefits at the same time. A common application portal may alter participation in several programs with different benefit rules, administrative match rates, and longer-run fiscal consequences. Moreover, the federal share that matters is the share for each marginal component of the policy. An average match rate for the program does not tell us whether the particular new activity qualified for reimbursement, whether the relevant grant was capped, or whether an administrative saving could actually be removed from the agency's budget.

I formalize these distinctions using three accounting boundaries. Let W_N denote the welfare value included in the consolidated calculation, and let W_s be the portion accruing to residents of state s. Let $G_{SL,s}$ be the net cost to the state and its local governments, $G_{F,s}$ the associated federal net cost, and $G_{O,j}$ an included cost in another government or outside account. I write costs as positive numbers and savings as negative numbers. The three objects are

$$M_N = \frac{W_N}{G_{SL,s} + G_{F,s} + \sum_j G_{O,j}}, \qquad M_{T,s} = \frac{W_s}{G_{SL,s}}, \qquad M_{R,s} = \frac{W_s}{G_{SL,s} + \lambda_s G_{F,s} + \sum_j \omega_{sj} G_{O,j}}. \tag{1}$$

The first ratio is the conventional MVPF, with government accounts consolidated. The second takes the limit of the state and local treasury's budget and excludes fiscal costs recorded elsewhere. I refer to the third as the state-resident MVPF. The parameter λ_s is the share of a marginal federal fiscal dollar borne by residents of state s, while ω_{sj} assigns the analogous incidence for any other outside component. If these weights are all set to one and the underlying ledgers are the same, the denominator is again the consolidated denominator. The full conventional MVPF is recovered when the resident and national welfare numerators coincide as well.

It is useful to consider a one-dollar increase in federally financed SNAP benefits paid to a resident. The treasury denominator omits the dollar because it never appears in the state budget. The state-resident denominator counts λ_s dollars, reflecting the resident share of federal financing, and the consolidated calculation counts the whole dollar. Across the three accounts, the treatment effect and the benefit received are the same. What changes is how much of the fiscal cost enters the denominator.

For λ_s, I report two transparent reference values: the state's population share and its share of federal individual income taxes after credits. Neither value estimates the incidence of the marginal federal dollar. The federal government could finance that dollar through payroll taxes, corporate taxes, borrowing, or a reduction in some other expenditure. I use these two values because their construction can be observed and because the comparison reveals whether a result turns on the choice between two plausible measures of state size. If included welfare is positive and the corresponding denominator is zero or negative, I report that the policy pays for itself under the included components. Reporting the ratio in this case would be misleading. A negative value would attach an unfavorable sign to a fiscal saving, and a denominator close to zero would produce an arbitrarily large number.

The empirical exercise reallocates published welfare accounts; it does not reestimate the causal effects of the policies. I preserve each paper's treatment effect and its convention for willingness to pay wherever that is possible. I then assign benefit spending, administrative costs, fiscal externalities, and implementation resources to their respective payers when the paper or contemporaneous program evidence provides enough information. Where the payer cannot be identified, I report scenarios.

Throughout the paper, I use four labels to make clear which parts of these calculations come directly from the evidence. An input is *verified* if it is reported in the paper or in an authoritative program source. It is *derived* if it follows arithmetically from those inputs, and *illustrative* if a missing incidence quantity requires an explicit assumption. I call an input *open* when the public evidence does not identify it. These labels describe individual inputs and calculations. They are not assessments of an entire paper.

This exercise is related to the MVPF comparisons in Hendren and Sprung-Keyser (2020). Their conventional calculation consolidates government accounts so that moving a dollar between them creates no social value on its own. I retain this object as the national benchmark. I then ask how the same fiscal components enter the decision of a government representing a narrower constituency. The change in the resulting MVPF comes from reallocating the existing denominator by fiscal incidence and restricting the numerator to welfare accruing to residents. No new welfare effect is generated by changing the boundary. The state-resident value is therefore useful for decentralized policy choice, while the national value remains the relevant one for questions about national efficiency.

Notice that fiscal reallocation leaves several of the difficult questions in take-up analysis exactly where they were. It cannot tell us whether a nonparticipant lacked information, whether an application burden used real resources, or how a marginal recipient valued an in-kind transfer relative to its budgetary cost. A large treasury ratio also provides no general verdict about the national desirability of the policy.

I begin with Pennsylvania, where a randomized change in participation gives a finite benchmark. I next consider California, where the assumed administrative savings can reverse the sign of the denominator. The third main case is Virginia's CommonHelp portal. Its several programs and payers support a set of informative scenarios, although the available evidence does not deliver a unique state-resident estimate. I close the empirical discussion with shorter cases in which match rates, the relevant policy boundary, or implementation costs remain unresolved.

2. Pennsylvania SNAP: a finite benchmark

The Pennsylvania experiment is a useful place to begin because both the policy and the fiscal accounting are relatively simple. Finkelstein and Notowidigdo (2019) studied Medicaid recipients age 60 or older who were not receiving SNAP and were thought likely to qualify. The state randomly assigned these recipients to a control group, an information treatment, or an information-plus-assistance treatment. Someone in the information group received a letter explaining that they might qualify for SNAP, a telephone number for the Pennsylvania Department of Human Services, and ordinarily a reminder postcard. Someone assigned to assistance received nearly the same information along with a referral to Benefits Data Trust (BDT). BDT screened callers, filled out and submitted forms, helped collect documents, and worked through problems that arose after submission.

The information treatment increased applications by 7.0 percentage points and enrollment by 4.7 points. Assistance increased applications by 16.1 points and enrollment by 11.8 points. Since only 5.8 percent of the control group enrolled, assistance produced an enrollment rate roughly three times as large as the control rate. It also changed the composition of SNAP recipients. The marginal enrollees were less sick and had higher incomes than existing recipients, and they were disproportionately likely to receive the $16 minimum benefit. Thus, valuing each induced dollar of benefits at one dollar would miss a feature of the experiment that matters for welfare: outreach drew in a different group of recipients.

Finkelstein and Notowidigdo infer that these marginal applicants had substantially underestimated the value of SNAP. In their behavioral calculation, the value of information is the gap between expected benefits and the value implied by a person's previous decision to forgo applying. The assistance calculation also includes the reduction in private application burden. Applying the paper's type-specific belief errors, benefit amounts, and application effects yields welfare of $199.152 per targeted person in the information arm and $466.8384 in the assistance arm. Of the latter amount, $17.85 comes from assuming that assistance eliminates a $75 application cost for each of the 23.8 percent of targeted people who apply.

Costs are calculated for the same population of targeted recipients. Information induces $205.20 of expected benefits and $18.69 of application processing, giving $223.89 before paying to deliver the treatment. The letter and postcard cost approximately $1, so total cost is $224.89 and the consolidated MVPF is 0.886. Assistance induces $462.78 of expected benefits and $42.72 of processing. If one assigns no resource cost to the assistance itself, the resulting ratio is 0.924. BDT's estimated cost is $7.20 per targeted person. Including that amount raises total cost to $512.70 and gives an MVPF of 0.911. I use this last figure for the implemented policy. BDT staff time is a real resource cost even when the state does not write the check.

To allocate these costs across governments, I use the financing rules in place in 2016. The federal government paid the full cost of SNAP benefits, while Pennsylvania and the federal government divided ordinary eligibility administration approximately equally. Available documents do not report whether the experiment obtained reimbursement for the mailings or the BDT services. I conservatively assign both costs to Pennsylvania. Under this assignment, the information arm costs Pennsylvania \$10.345 and the federal government \$214.545. The corresponding amounts for assistance are \$28.560 and \$484.140. Notice that there is no single federal share one can apply to the total. Benefits, routine processing, and delivery each have their own financing rule.

With these allocations, the resident calculations take a particularly simple form:

$$M^{I}_{R,PA}(\lambda) = \frac{199.152}{10.345 + 214.545\lambda}, \qquad M^{A}_{R,PA}(\lambda) = \frac{466.8384}{28.560 + 484.140\lambda}. \tag{2}$$

Pennsylvania accounted for 3.956 percent of the US population in 2016, so setting $\lambda = 0.03956$ gives 10.57 for information and 9.78 for assistance. Using Pennsylvania's share of federal individual income tax after credits produces 10.68 and 9.87. The two measures of federal incidence thus lead to very similar answers in this case. At $\lambda = 0$, which corresponds to the state treasury limit, the ratios are 19.25 and 16.35. At $\lambda = 1$, they return to the consolidated values of 0.886 and 0.911. All of the state figures are my calculations under the financing and incidence assumptions described above. Finkelstein and Notowidigdo do not report them.

The treatment-delivery assignment matters, although it does not account for most of the gap between the state and consolidated values. If routine processing and the experimental outreach or BDT resources are each divided 50/50, the population-share estimates are 10.85 for information and 10.55 for assistance. Assigning Pennsylvania all processing and delivery costs lowers them to 7.16 and 6.84. Every one of these denominators is positive. Appendix B keeps the alternatives separate because the fact that an activity was eligible for reimbursement does not establish that reimbursement was requested or received.

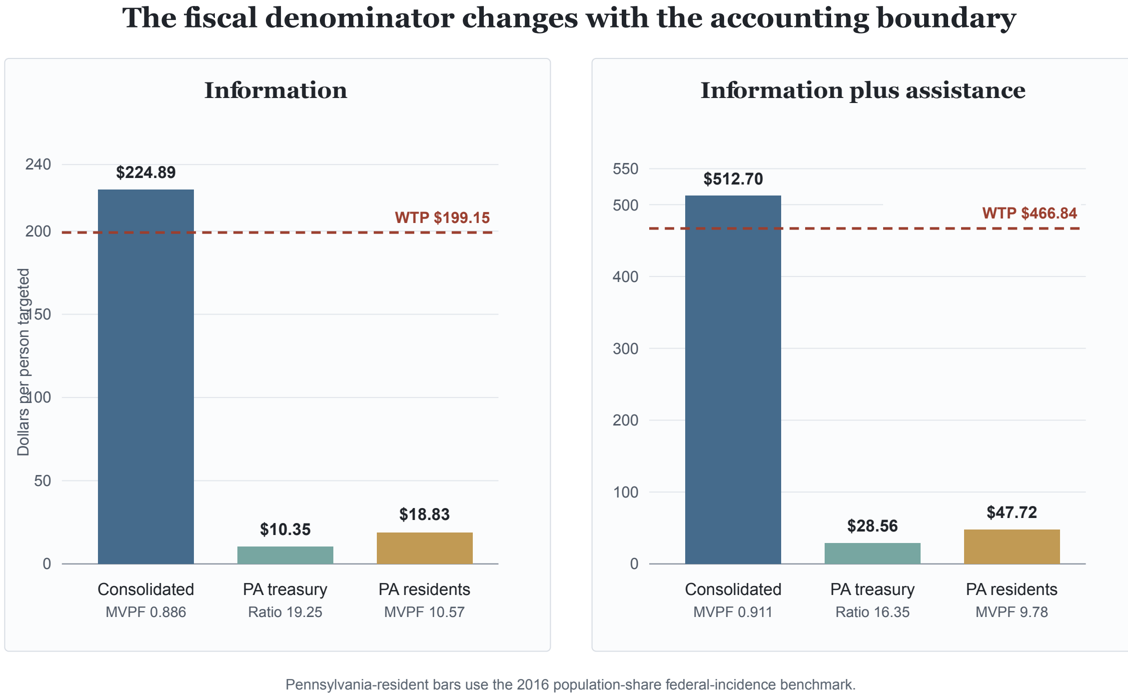

Figure 1. Each panel holds recipient willingness to pay fixed and changes only the fiscal boundary. The Pennsylvania-resident bars use the state's 2016 population share as the federal-incidence weight. Dollar amounts are per person targeted. Source: author's calculations from Finkelstein and Notowidigdo (2019) and its online appendix.

What causes an MVPF below one to become a value near ten? Consider the information treatment. Its $199.152 of welfare accrues to people in Pennsylvania. The resident denominator includes the full Pennsylvania cost and only $8.49 of the $214.545 federal cost, since Pennsylvania residents finance only a small share of federal spending. This gives a denominator of $18.83. For assistance, the same calculation produces a denominator of $47.72 and an MVPF of 9.78. Assistance generates considerably more enrollment, but it also uses more processing and BDT resources that I have assigned to Pennsylvania. Its state-resident value is consequently a little lower.

Both denominators remain positive. Unmeasured health or tax savings would have to exceed $18.83 for information or $47.72 for assistance before either policy paid for itself on the included components. Fixed costs for management, data, call-center capacity, or IT would move the denominator in the other direction. The paper does not provide a policy-specific value for any of these items, and the identity of the payer remains open.

There is substantial uncertainty in the numerator as well. The randomized experiment identifies the enrollment effects, while prior beliefs and willingness to pay are inferred. The assumed $75 burden reduction in the assistance arm comes from five hours valued at $15 per hour, and recipients still had to complete the required state interview. There is also an unresolved discrepancy in the cost calculation. The $267 processing-cost input reproduces the published welfare calculation, although a nearby administrative statistic is $134 and the two figures do not transparently reconcile. The burden assumption enters willingness to pay, while the processing and implementation costs enter the fiscal denominator. Neither changes the randomized participation effect.

3. California CalFresh: when simplification saves money

California adopted semiannual reporting for CalFresh in October 2013. Most households had previously submitted an eligibility report every quarter, and the reform eliminated the report due around the third month after enrollment. Unrath (2024) compares cohorts entering just before and just after the change. Six-month retention increases from 78.3 to 87.0 percent, or 8.7 percentage points. The evidence here comes from an adjacent-cohort comparison. The paper does not display an untreated group or sampling uncertainty, and its text and figure note refer to different pre- and post-reform windows. These features make the participation estimate less secure than the randomized estimates from Pennsylvania.

In the March 2024 version of the paper, the consolidated MVPF is 4.25. The calculation is normalized to one high-food-security case and one low-food-security case. Over months four through six, the reform generates \$138.16 in additional benefits. It also removes \$42.02 of private reporting burden for marginal and inframarginal recipients, giving total willingness to pay of \$180.18. Government pays the \$138.16 in benefits and incurs a \$9.26 fiscal externality drawn from other studies. It is also assumed to avoid \$105.05 in processing costs. The resulting government cost is \$42.37, and $180.18/42.37 = 4.25$.

The historical financing rules place each of these fiscal components in a different account. The federal government financed CalFresh benefits. Ordinary administration was 50 percent federal, 35 percent state, and 15 percent county. The \$9.26 fiscal externality combines an assumed adult labor-supply cost with a future child-earnings saving, both imported from other studies. Neither Unrath nor the underlying papers identifies how this amount should be divided across governments. I therefore let q_K denote the share assigned to California state and local governments, and let λ denote the share of federal cost borne by California residents. The resident denominator is

$$D_{CA}(\lambda, q_K) = -52.525 + 9.26q_K + \lambda\,(94.895 - 9.26q_K)\,. \tag{3}$$

The term -52.525 is the state and county share of the administrative saving assumed by the paper. The next term assigns part of the fiscal externality to California. Inside the parentheses are the federal benefits, the federal administrative savings, and whatever part of the externality is assigned to the federal government. I leave q_K in the expression because the available evidence provides no basis for choosing one allocation.

Suppose first that none of the fiscal externality falls on California. State and local governments then save \$52.53 per normalized policy unit. If all of it falls on California, they save \$43.27. Adding the federal cost attributable to California residents does not change the sign under either incidence measure. The resident denominator ranges from -\$41.02 to -\$32.88 when λ is based on population and from -\$39.44 to -\$31.46 when it is based on federal income tax after credits. Thus, under the paper's component assumptions and the stated financing and incidence scenarios, the policy pays for itself from the perspective of California residents. Reporting a negative MVPF would be misleading because it would attach a negative number to a policy that generates positive welfare and fiscal savings.

The assumed administrative saving is doing considerable work in this calculation. Unrath assigns a public cost of \$50 to every report and applies the saving to both marginal and inframarginal cases. The paper does not estimate the marginal cost of processing a report in California. It also does not show how much of the freed worker time eventually appeared as a reduction in county or state spending. Consider the combination of q_K and λ that produces the largest resident cost. The denominator reaches zero if 47.4 percent of the assumed \$105.05 saving is marginal and cashable, in the sense that it eventually reduces public spending. Any realized share above 47.4 percent preserves the negative denominator. Whether the actual share clears this threshold is an empirical question that the available evidence cannot answer.

Transition costs are missing from the calculation as well. The reform required systems work, notices, and training. State fiscal documents establish that these activities occurred, but the reported totals combine several programs and provisions and cannot be mapped into Unrath's two-case normalization. Several other inputs are calibrations. The \$20 private reporting cost does not follow from the time-and-wage calculation printed next to it, and the adult and child fiscal effects come from national studies of SNAP receipt. They were not estimated from the California reporting reform.

One can still ask how large the missing transition cost would need to be. If California paid all of it, a cost between \$31.46 and \$41.02 per normalized unit would exhaust the resident fiscal saving across the cases considered above. With an equal federal and California split, the corresponding range is \$55.29 to \$73.17. The aggregate fiscal notes omit the exposure and amortization terms required to put their statewide totals on this scale. For this reason, these figures are best read as the transition-cost amounts that would reverse the sign, conditional on the rest of the calculation.

4. Virginia CommonHelp: several programs and several payers

Virginia's CommonHelp portal changed access to several programs at once when Virginia opened it statewide in October 2012. Households could use the portal to apply online for SNAP, Medicaid and FAMIS, TANF, child-care assistance, and energy assistance, and online recertification was added later. For households with children, Cholli and Wu (2026) use distance to local social-service offices and discontinuities at office-assignment boundaries to estimate how participation responded to the reduction in administrative burden. They find that multiple-program participation rose by 4.183 percent and single-program participation fell by 1.732 percent. Since the eligible population for each combination of programs is unavailable, I interpret these estimates as changes in the composition of the caseload. Identification of take-up among verified eligible households would require this additional denominator.

For the complete CommonHelp policy, Cholli and Wu report a consolidated MVPF of 1.27. Their welfare calculation includes time and travel saved during application and recertification, the value of additional program receipt, the \$10.78 million cost of developing the portal, changes in benefits and variable administration, and several fiscal externalities. One of the larger externalities assumes one dollar of long-run government medical savings for every dollar spent on childhood Medicaid. The paper also reports separate calculations for Medicaid and SNAP, although recovering the complete-policy estimate requires a joint calculation. CommonHelp changed which combinations of programs households received, using one portal to change access to all of them.

How much of the resulting cost belongs to Virginia? CommonHelp spans several program-specific financing rules. SNAP benefits were financed federally, while the cost of administering SNAP was shared. Medicaid and FAMIS had different match rules, and the composition of marginal child enrollment between the two programs is unavailable. TANF was financed with a fixed block grant and state maintenance-of-effort spending. For TANF, an accounting share may have little connection to the opportunity cost of one more dollar of spending. Virginia and its local departments divided some administrative responsibilities, and the portal itself was part of a larger eligibility system financed from several sources. Long-run medical savings present an additional problem, since the government receiving those savings depends on where a child later lives, what coverage they have, and which public payer pays for that coverage.

A complete numerical decomposition is unavailable in the public paper. Except for development spending, the inputs appear only as bars in Figure A.32. The paper supplies no numerical workbook or replication code for the welfare exercise. I use approximate graph-calibrated anchors of $63.10 million for total welfare value and $85.30 million for direct government cost. I then recover a fiscal externality of -$35.615 million as the residual needed to reproduce the authors' exact MVPF of 1.27, giving a consolidated net cost of $49.685 million. Cholli and Wu report the consolidated calculation, while the component anchors and state calculations below are my derived scenarios calibrated to Figure A.32.

For my benchmark, I allocate $12.758 million of net cost to Virginia state and local accounts and $36.927 million to a combined federal and outside-government account. I use historical program rules where they are available and state the remaining assumptions. In particular, the benchmark assigns none of the modeled $37.1 million child-Medicaid saving to Virginia. It divides portal development equally between Virginia and federal accounts and assigns 20 percent of the small remaining tax-and-transfer externality to Virginia. With Virginia's population share as the incidence weight, the resident denominator is $13.723 million. The complete-policy MVPF is 1.27 under consolidated accounting, 4.95 at the treasury limit, and 4.60 for Virginia residents. Using Virginia's share of federal income tax produces 4.56.

I therefore treat 4.60 as a benchmark and consider alternative allocations separately. The calculation applies one λ to a combined federal and outside-government account, although the payers collected in that account are quite different. A direct-budget sensitivity omits all modeled behavioral fiscal externalities and gives 4.39. This sensitivity uses a different component ledger, since the fiscal externalities have been removed. The benchmark's treatment of the long-run medical saving tends to lower the Virginia-resident ratio. The direction of the remaining sensitivities depends on the component, since changing the TANF opportunity cost, Virginia's share of portal development, or the division of marginal enrollment between Medicaid and FAMIS could raise or lower the ratio.

A later report on Virginia's broader eligibility system assigns 18.7 percent of development spending to the state general fund. If I use that share for CommonHelp and treat the TANF residual as fully controlled by Virginia, the resident ratio rises to about 7.26 and the medical-saving crossing falls to about 24 percent. The later report covers more than the 2012 portal and comes from a different period, so I leave this calculation as a sensitivity. It suggests that a 50/50 division of portal development may assign too much of that cost to Virginia. The marginal financing of the original CommonHelp portal remains unknown.

The largest source of variation is the jurisdiction of the long-run medical saving. Let s denote the share of the modeled \$37.1 million saving that accrues to Virginia, with the remaining share assigned to the combined federal and outside-government account. Under the population-share incidence benchmark, the resident denominator is

$$D_{VA}(s) = 13.723 - 37.1(1 - 0.02613)s. \tag{4}$$

The factor $1 - 0.02613$ accounts for the fact that a dollar assigned to Virginia enters the resident denominator in full. If that dollar remains in the combined federal and outside-government account, it already enters the denominator with Virginia's population-share weight. Assigning one-quarter of the saving to Virginia lowers the resident denominator to \$4.690 million and raises the ratio to 13.45. The denominator reaches zero at approximately $s = 0.38$. For larger Virginia shares, the policy pays for itself under this scenario.

The 38 percent threshold answers a question about fiscal jurisdiction. Given the modeled saving, it is the share that would have to accrue to Virginia for the resident denominator to reach zero. The threshold carries no prediction that Virginia will actually receive this amount. The underlying evidence follows children in a different setting and extrapolates effects on medical utilization through age 65, while future residence and the identity of the public payer remain unknown. I therefore carry forward two results from CommonHelp: the benchmark of 4.60 and the crossing at 38 percent.

5. Additional participation cases

The four remaining studies each leave at least one input needed for a state calculation unresolved. Depending on the study, the missing input is the match rate, the boundary of the policy being evaluated, or implementation cost. Three studies evaluate rules that directly change enrollment or retention, while a fourth estimates how caseworkers affect participation without evaluating an intervention that would produce the observed change in caseworker effectiveness. The component calculations appear in Appendix E.

Shepard and Wagner (2025) study automatic assignment to a health plan in Massachusetts Commonwealth Care. Applicants with income below poverty who had already been found eligible were assigned to a plan if they failed to choose one. When automatic enrollment was suspended, new enrollment fell by 32.6 percent relative to the automatic-enrollment baseline. Equivalently, adding automatic enrollment increased new enrollment by 48.4 percent relative to active enrollment alone. The paper's consolidated welfare ratio is approximately one.

To construct a Massachusetts-resident calculation, I assign 36.5 percent of uncompensated care to government and the remaining 63.5 percent to providers. I assume that the government offset receives the same match as insurance coverage. The ownership of the affected providers and the residence of their owners are unavailable, so I allow the share of the provider benefit accruing to Massachusetts residents to range from zero to one. I use a 3.228 percent federal-incidence weight and set unmeasured implementation cost to zero. Under these assumptions, the resident ratio ranges from 1.01 to 1.94 with a 50 percent match. At a 61.59 percent match, it ranges from 1.29 to 2.48. I report these ranges because enrolled months span different financing regimes, the marginal public offset for uncompensated care is unknown, and some of the benefit accrues to providers whose residence and ownership cannot be observed.

Gray et al. (2023) examine Virginia's historical three-month time limit for able-bodied adults without dependents. Eliminating the time limit increases participation, although it also changes substantive eligibility and work conditions in addition to recertification burden. The authors report a consolidated MVPF between 0.90 and 1.40. Under the historical financing rules, elimination saves the Commonwealth and local governments $1.21 per relevant ABAWD-month on the measured components.

For the resident calculation, I use the authors' central earnings calibration over 24 months and assume zero net Virginia income tax for the representative filer. I apply the published WTP endpoints and use both population and federal-income-tax shares to allocate federal costs. The resulting resident MVPFs range from 64.56 to 118.08. The denominators producing those values are only about $0.99 to $1.15. Thus, the large ratios mainly reflect the fact that the denominator is close to zero. A modest transition cost omitted from the calculation, or a change in the earnings calibration, can be more consequential than the numerical difference between 65 and 118.

Cook and East (2026) study parents who were already receiving SNAP in an anonymous state. When the youngest child turned six shortly before recertification, the household head could be referred to mandatory Employment and Training. The authors consider a counterfactual that eliminates mandatory E&T, restoring an estimated $111.50 in monthly SNAP benefits per induced referral and subtracting $11.32 in planned average E&T cost. Its consolidated MVPF is 1.11.

The state-resident calculation depends on how E&T was financed, since funding could come from a finite grant with a 100 percent federal share, from 50/50 reimbursement, or from an unidentified nonfederal payer. Because the marginal financing source remains unidentified, a state-resident point cannot be recovered from the paper. For an illustration, I use a federal-incidence weight of 0.02 and treat all supervised job search as formula-funded. I assume that released formula dollars reduce federal outlays dollar for dollar, allocate all nonfederal reimbursement savings to state residents, and set omitted fiscal and transition effects to zero. This produces a resident value of 103.95. If the same activity is treated as 50/50 funded, the denominator can become nonpositive. The sign change is more informative than a range whose upper endpoint becomes arbitrarily large as the denominator approaches zero.

A separate paper by Cook and East (2025) uses quasi-random assignment to SNAP caseworkers in an anonymous mountain-plains state. Assignment to a caseworker who is one standard deviation more effective at producing receipt raises initial receipt by 2.68 percentage points and reduces incomplete applications by 1.27 points. The authors report a hypothetical welfare calculation for a one-standard-deviation improvement in effectiveness. They report 3.3, while their unrounded displayed inputs produce 3.13.

This paper provides strong evidence that caseworkers influence what happens after an application has been submitted. The policy that would make caseworkers more effective is left unspecified, and training, staffing, monitoring, caseload changes, and software could each change worker performance at a different cost. The paper measures none of those technologies. Since the implementation technology, its cost, and the state are unidentified, an implementable state-resident MVPF remains unavailable.

6. Lessons for state policy

A state deciding whether to reduce a participation barrier needs three pieces of evidence. First, it needs an estimate of how much participation changes, which Pennsylvania provides through random assignment. California relies on a less secure adjacent-cohort comparison, while the CommonHelp study uses geographic and office-boundary designs. The state also needs to know how the marginal participants value the change, which requires a view about misinformation, application ordeals, the value of transfers, or future benefits. Finally, it needs to know who bears the fiscal consequences, including how federal costs ultimately reach state residents. These are separate empirical exercises. A strong causal design can leave the payer unknown, and a high federal match by itself says nothing about how recipients value the benefit.

The fiscal calculations also show why the name of the program tells us relatively little about the relevant cost. In Pennsylvania, the dominant fact is that induced benefits are federally financed, while processing is divided 50/50 and the delivery resources are conservatively assigned to Pennsylvania. In California, the resident denominator changes sign because of the assumed state and county administrative savings. CommonHelp brings together open-ended benefits, matching administration, a fixed block grant, portal development, and long-run medical savings that may be received by another jurisdiction. Each component therefore requires its own marginal financing analysis. An expenditure that is eligible for reimbursement may never have been claimed. An average federal share can be uninformative when a grant cap binds, and one less hour of worker time saves cash only when it reduces staffing or another paid input. Statutory average shares provide a starting point and leave these marginal questions open.

I therefore regard the component ledger as at least as important as the final ratio. The ledger should report resident willingness to pay, state and local net cost, federal net cost, any outside fiscal component, and the incidence weights used to combine them. A scenario range records what happens when an unobserved payer or a normative parameter is changed. Sampling uncertainty concerns the precision with which an underlying effect was estimated. Keeping these two ranges separate matters whenever the denominator depends on an unobserved allocation.

Notice, this issue becomes especially important when the denominator is close to zero. A one-dollar change can turn a large finite MVPF into a fiscal saving. Reporting both the denominator and the value at which it crosses zero helps the reader understand why an MVPF of 100 built on one dollar of cost may be less stable than an MVPF of ten built on fifty dollars. Implementation, transition, and IT costs also belong in the ledger, with explicit bounds showing how large these costs would have to be to change the result whenever direct estimates are unavailable.

What should a state take from these calculations? Take-up policies may be considerably more attractive from the perspective of the state treasury and state residents than their consolidated-government MVPFs suggest. The Pennsylvania interventions produce finite resident values near ten because most of the additional benefits are paid by the federal government. The California reform pays for itself under the included components when enough of the assumed administrative saving can be realized. For CommonHelp, the benchmark resident MVPF is 4.60 based on a reconstruction of Figure A.32, and the result is quite sensitive to where future medical savings accrue.

Because these findings depend on the particular policy and its financing, a useful welfare calculation should report components by level of government, document the reimbursement claims that were actually made, measure marginal administrative and implementation costs, and identify the jurisdiction receiving each fiscal externality. When one of these inputs is unavailable, stating the missing input and reporting the associated crossing or range gives the state a more useful answer than folding everything into a consolidated ratio.

Appendix A. Accounting definitions and conventions

A.1 The three accounting boundaries

For a policy implemented in state s, let W_N be the welfare numerator in the national calculation and let W_s be the portion accruing to residents of that state. I use $G_{SL,s}$, $G_{F,s}$, and $G_{O,j}$ for the net fiscal costs borne by state and local government, the federal government, and an outside account j. Costs are positive and savings are negative. With this notation, the three objects in Equation (1) are

$$M_N = \frac{W_N}{G_{SL,s} + G_{F,s} + \sum_j G_{O,j}}, \quad M_{T,s} = \frac{W_s}{G_{SL,s}}, \quad M_{R,s} = \frac{W_s}{G_{SL,s} + \lambda_s G_{F,s} + \sum_j \omega_{sj} G_{O,j}}.$$

The treasury-limit ratio asks how much welfare accrues to state residents per dollar of state and local budget cost. Fiscal accounts elsewhere do not enter this denominator, so the ratio should be read as a welfare-over-budget statistic. The state-resident calculation then adds the shares of outside costs that ultimately fall on residents. In the empirical work, I vary λ_s using population and federal individual-income-tax shares. I specify ω_{sj} separately when an outside component has an incidence that differs from federal spending.

The three ratios have a useful relationship in the special case with one state and local account and one federal account. Suppose they have the same numerator and these two accounts exhaust the denominator. Since

$$G_{SL,s} + \lambda G_{F,s} = (1-\lambda) G_{SL,s} + \lambda (G_{SL,s} + G_{F,s}),$$

it follows that

$$\frac{1}{M_{R,s}(\lambda)} = \frac{1-\lambda}{M_{T,s}} + \frac{\lambda}{M_N}. \tag{A.1}$$

Thus, the inverse resident MVPF is a weighted average of the inverse treasury and national ratios. Notice that Equation (A.1) requires a common numerator and a fixed, exhaustive two-account ledger. It ceases to describe the relationship once the resident and national numerators differ, an outside account has been omitted, or a sensitivity changes the components themselves. The CommonHelp direct-budget case removes behavioral fiscal externalities, for example. It consequently defines a different ledger and falls outside this interpolation.

A.2 Incidence, signs, and evidence labels

Population shares and income-tax shares provide two transparent rules for allocating a federal cost across states. Neither traces the source of financing for a marginal federal dollar. That dollar could come from current income, payroll, corporate, or excise taxes, from borrowing and future taxes, or from reduced federal spending elsewhere. I use these benchmarks because they are dated and reproducible, and because showing both reveals how much a result depends on the particular measure of state size.

If $W_s > 0$ and the resident or treasury denominator is nonpositive, I say that the policy **pays for itself under the included components and scenario**. A negative or infinite MVPF has little useful interpretation here. When a positive denominator approaches zero, I give both its level and the parameter value at which it crosses zero. This makes clear when a very large ratio is being generated by a denominator close to zero.

I use four evidence labels throughout the appendices:

- **Verified**: printed in the relevant paper or an authoritative program, budget, statutory, or statistical source.
- **Derived**: arithmetic using verified inputs.
- **Illustrative**: a stated value used because an incidence, valuation, or implementation input is unavailable.
- **Open**: not identified by the reviewed public evidence and not silently assigned.

The labels apply to individual claims. Thus, a causal estimate may be verified even though its fiscal allocation is illustrative. Similarly, a statutory financing share may be verified while the amount actually reimbursed in this implementation remains open.

Appendix B. Pennsylvania SNAP component ledgers

B.1 Reconstruction of the two treatments

All amounts in this appendix are dollars per person targeted. In the information arm, the application effects for the high- and low-benefit groups are 0.04 and 0.03. The prose in Finkelstein and Notowidigdo (2019) reverses the subscripts, while its displayed equation and reported composition of marginal applicants use the ordering below. I follow the displayed calculation:

$$W_I = .98(4{,}806)(.04) + .83(432)(.03) = 199.152.$$

For assistance, the corresponding high- and low-benefit effects are 0.09 and 0.07. These effects imply a behavioral value of \$448.9884 before accounting for application burdens. The source calculation adds \$75 for each of the 23.8 percent of treated people who apply:

$$W_A = .98(4{,}806)(.09) + .83(432)(.07) + 75(.238) = 466.8384.$$

The following table gives the complete ledger for each treatment. The national column reconstructs the conventional calculation. The Pennsylvania column shows the allocation I use for the state-resident calculation.

Treatment and component	Amount	National ledger	Recommended Pennsylvania allocation	Status
Information: recipient WTP	199.152	numerator	resident numerator	Derived from printed inputs
Information: induced benefits	205.200	cost	205.200 federal	Derived; benefit financing verified
Information: processing	18.690	cost	9.345 state, 9.345 federal	Derived; historical 50/50 benchmark
Information: mail and reminder	1.000	resource cost	1.000 state	Approximate amount; payer open, allocation illustrative
Information total	**224.890**	**denominator**	**10.345 state, 214.545 federal**	Derived
Assistance: belief-based WTP	448.988	numerator	resident numerator	Derived from printed inputs
Assistance: burden relief	17.850	numerator	resident numerator	Illustrative calibration
Assistance: induced benefits	462.780	cost	462.780 federal	Derived; benefit financing verified
Assistance: processing	42.720	cost	21.360 state, 21.360 federal	Derived; historical 50/50 benchmark

Treatment and component	Amount	National ledger	Recommended Pennsylvania allocation	Status
Assistance: BDT resources	7.200	resource cost	7.200 state	Derived amount; payer open, allocation illustrative
Assistance total	**512.700**	**denominator**	**28.560 state, 484.140 federal**	Derived

To connect the table to the published estimates, remove the mailing resource from the information denominator. This gives 199.152/223.890 = 0.8895, which is the published 0.89. Removing BDT resources from the assistance denominator gives 466.8384/505.500 = 0.9235. The original appendix reports 0.93 and the secondary reconstruction reports 0.92. Once the implementation resources are included, the two ratios are 0.886 and 0.911.

The current Policy Impacts information page displays dollar fields that fail to reproduce the formula on the page, although the resulting ratio still rounds to 0.89. Its assistance page reports a normalized value near 0.92 without showing whether assistance is treated as costless or as a resource cost. I therefore reconstruct both treatments from the article and its online appendix.

One input remains unresolved. The public calculation assigns $267 of processing cost to each application, and this number reproduces the welfare equation. Nearby, the article describes a cost of approximately $134 per application, obtained by multiplying $178 per recipient by a 75 percent acceptance rate. The article and online appendix do not explain how the $267 and $134 figures relate. I retain $267 because it reconstructs the authors' calculation, while treating the provenance of this input as open.

B.2 Federal incidence and financing alternatives

Applying the recommended allocations from Table B.1 gives

$$M^{I}_{R,PA}(\lambda) = \frac{199.152}{10.345 + 214.545\lambda}, \qquad M^{A}_{R,PA}(\lambda) = \frac{466.8384}{28.560 + 484.140\lambda}.$$

The table traces both ratios as the federal-incidence weight increases from the treasury limit to the consolidated calculation.

Federal-incidence weight λ	Information denominator	Information ratio	Assistance denominator	Assistance ratio
0, treasury limit	10.345	19.251	28.560	16.346
0.020	14.636	13.607	38.243	12.207
0.038705, income-tax share	18.649	10.679	47.299	9.870

Federal-incidence weight λ	Information denominator	Information ratio	Assistance denominator	Assistance ratio
0.039564, population share	18.833	10.574	47.715	9.784
0.050	21.072	9.451	52.767	8.847
0.100	31.800	6.263	76.974	6.065
0.250	63.981	3.113	149.595	3.121
0.500	117.618	1.693	270.630	1.725
1, consolidated	224.890	0.886	512.700	0.911

The population benchmark divides Pennsylvania's July 2016 population, 12,784,227, by the U.S. population, 323,127,513. The income-tax benchmark divides Pennsylvania's 2016 federal individual income tax after credits, $55.697 billion, by the national total of $1,439.016 billion. Both remain illustrative incidence proxies, since each allocates the federal cost according to an observed state share.

There is a separate question about the experimental delivery resources. The available evidence does not show which government ultimately paid for the mailing, reminder, and BDT costs. The next table holds welfare and consolidated resource cost fixed and changes only the allocation of those costs.

Allocation	State cost	Federal cost	Population-share ratio	Treasury ratio
Information: processing 50/50, outreach all state	10.345	214.545	10.57	19.25
Information: processing and outreach 50/50	9.845	215.045	10.85	20.23
Information: processing and outreach all state	19.690	205.200	7.16	10.11
Assistance: processing 50/50, BDT all state	28.560	484.140	9.78	16.35

Allocation	State cost	Federal cost	Population-share ratio	Treasury ratio
Assistance: processing and BDT 50/50	24.960	487.740	10.55	18.70
Assistance: processing and BDT all state	49.920	462.780	6.84	9.35

USDA rules permitted federal administrative reimbursement for approved factual outreach and application assistance. This establishes that the expenses could qualify for a match. The public evidence does not show whether Pennsylvania submitted these experimental expenses for reimbursement or received a federal payment. I use the all-state allocation as a conservative upper bound on the state denominator; the actual invoicing remains open.

B.3 WTP and omitted-cost sensitivities

The \$75 value assigned to application assistance is an important valuation choice. Let r denote the burden reduction per applicant. The assistance numerator can then be written as

$$W_A(r) = 448.9884 + .238r,$$

Holding the recommended population-share denominator fixed gives the following sensitivity.

WTP treatment	Information WTP	Information ratio	Assistance WTP	Assistance ratio	Status
Authors' behavioral baseline; full \$75 assistance relief	199.152	10.57	466.8384	9.78	Source calibration plus derived reallocation
No application-burden relief	199.152	10.57	448.988	9.41	Illustrative
Relief only for induced applications, $.16 \times 75$	199.152	10.57	460.988	9.66	Illustrative

WTP treatment	Information WTP	Information ratio	Assistance WTP	Assistance ratio	Status
Mechanical value equal to induced benefits; full assistance relief	205.200	10.90	480.630	10.07	Illustrative

The calculation may also omit fixed, transition, management, data, call-center, or IT costs. Let k be the amount of any such omitted cost per targeted person paid by Pennsylvania. The resident formulas become

$$\frac{199.152}{10.345 + k + 214.545\lambda} \quad \text{and} \quad \frac{466.8384}{28.560 + k + 484.140\lambda}.$$

At the population benchmark, fiscal savings of \$18.833 per information target or \$47.715 per assistance target would bring the corresponding denominator to zero. The experiment does not estimate a health, tax, or other saving of this magnitude. The state-resident ratios exceed one for $\lambda < 0.8800$ and $\lambda < 0.9053$, respectively. Both denominators remain positive when the calculation is restricted to the measured components.

Finally, these allocations describe the 2016 experiment. Later changes in SNAP administrative or benefit cost sharing do not update the experiment by themselves. A prospective calculation would require a new take-up response, delivery technology, state payment-error status, and reimbursement treatment.

Appendix C. California CalFresh component ledger and thresholds

C.1 Reconstructing the March 2024 calculation

Unrath's Figure 13 is the only internally consistent reconstruction of the reported 4.25:

Component	Amount	Role	Fiscal allocation in the historical benchmark
Additional benefits, high-food-security type	71.94	WTP and cost	federal
Additional benefits, low-food-security type	66.22	WTP and cost	federal

Component	Amount	Role	Fiscal allocation in the historical benchmark
Inframarginal private burden avoided	40.00	resident WTP	California recipients
Marginal private burden avoided	2.02	resident WTP	California recipients
Adult fiscal costs	14.41	government cost	jurisdiction open
Child future-earnings fiscal savings	-5.15	government saving	jurisdiction open
Inframarginal public administration avoided	-100.00	government saving	50 federal, 35 state, 15 county
Marginal public administration avoided	-5.05	government saving	2.525 federal, 1.7675 state, .7575 county
Total	**WTP 180.18; net cost 42.37**	**MVPF 4.25**	n/a

Set benefits at $B = 138.16$, private burden relief at $H = 42.02$, the net transported fiscal externality at $K = 9.26$, and administrative savings at $A = 105.05$. Then

$$W = B + H = 180.18, \qquad G = B + K - A = 42.37, \qquad M_N = 4.2525.$$

The compact equation on page 23 gives WTP of \$182.68, government cost of \$42.2364, and a ratio of 4.33. Its administrative multiplier is 2.14, while the components in Figure 13 sum to 2 + .054 + .047 = 2.101. The live Policy Impacts page reports an older calculation with WTP \$220.76, net cost \$105.19, and MVPF 2.09. I use Figure 13 in the March 2024 manuscript as the current authoritative ledger.

The private reporting cost contains a second discrepancy. The paper assigns \$20 to two hours valued at twice California's 2013 minimum wage. Since that wage was \$8, the stated formula gives \$32. I retain the operative welfare input of \$20.

C.2 Historical state and federal allocation

Let q_K be the share of the net \$9.26 fiscal externality assigned to California state and local government. During the period studied, benefits were entirely federal and ordinary administration was 50 percent federal, 35 percent state, and 15 percent county. Hence

$$G_{SL} = 9.26 q_K - 52.525, \qquad G_F = 94.895 - 9.26 q_K.$$

q_K	State/local cost	Federal cost	Resident cost, population $\lambda = .121256$	Resident cost, tax $\lambda = .137885$	Result
0	-52.53	94.90	-41.02	-39.44	pays for itself
0.5	-47.90	90.27	-36.95	-35.45	pays for itself
1	-43.27	85.64	-32.88	-31.46	pays for itself

Notice that every state and resident denominator is negative. The population weight divides California's 2013 population of 38,332,521 by the U.S. population of 316,128,839. The tax weight divides California's $169.422 billion in 2013 federal individual income tax after credits by the national $1,228.719 billion. Even the least favorable assignment, placing all adult cost in California and all child saving in the federal account, leaves resident costs of -$28.36 and -$27.02 under the two weights.

C.3 Realizable administration and transition costs

The central calculation treats all $105.05 of administrative savings as realizable. Let ρ_A be the fraction that is marginal, allowable, and cashable. The resident denominator becomes

$$D_{CA} = q_K K + \lambda[B + (1 - q_K)K] - .5(1 + \lambda)\rho_A A.$$

The minimum ρ_A that makes the denominator nonpositive is:

q_K	Treasury $\lambda = 0$	Population weight	Income-tax weight
0	0.000	0.304	0.340
0.5	0.088	0.373	0.407
1	0.176	0.442	**0.474**

The least favorable threshold in the central grid is 47.4 percent, though whether California could cash out that share remains unobserved. The paper imports a $50 processing cost from earlier work and contains no evidence on expenditure reductions or staff reassignment after the report was eliminated.

Let τ be transition, IT, training, notice, and rulemaking cost per normalized policy unit. With a 50/50 split between California and the federal government, the amount that brings the resident denominator to zero is:

q_K	Treasury threshold	Population threshold	Income-tax threshold
0	105.05	73.17	69.32

q_K	Treasury threshold	Population threshold	Income-tax threshold
0.5	95.79	65.91	62.31
1	86.53	58.65	55.29

If California bears the entire transition cost, the corresponding thresholds are \$52.53/\$47.90/\$43.27 for the treasury, \$41.02/\$36.95/\$32.88 with population weights, and \$39.44/\$35.45/\$31.46 with tax weights. Official fiscal documents report material statewide implementation amounts. Because the totals combine CalFresh with other provisions, converting them to the paper's normalized two-case unit requires an exposure count, a program allocation, and an amortization horizon.

These administrative and transition thresholds concern the denominator. Recipient valuation enters through the numerator. Valuing benefits at one-half of face value reduces WTP to \$111.10 and the national MVPF to 2.62. Valuing only private reporting relief gives WTP of \$42.02 and a national ratio of 0.99. Both sensitivities retain a negative resident denominator and positive WTP in the historical central grid.

Appendix D. Virginia CommonHelp reconstruction

D.1 Graph calibration

Cholli and Wu report an exact complete-policy MVPF of 1.27. Figure A.32 gives its dollar components as unlabeled bars. Reading the figure at 300 dpi gives approximately \$5.5 million of direct WTP, \$58.0 million of indirect WTP, \$85.7 million of direct government cost, and a -\$34.9 million fiscal externality. Given the coarseness of the plot, I use one-decimal anchors of \$63.10 million for total WTP and \$85.30 million for direct cost, then choose the fiscal externality that exactly reproduces 1.27:

$$FE^{all} = \frac{63.10}{1.27} - 85.30 = -35.614961.$$

Consolidated net cost is \$49.685039 million. The dollar components are derived from and calibrated to Figure A.32, while the 1.27 comes directly from the authors.

D.2 Program and payer ledger

The program-payer ledger below is constructed from the graph because Cholli and Wu report only aggregate component bars.

Component, millions of dollars	Amount	Benchmark Virginia/local share	Benchmark federal/outside share	Evidence and principal caveat
Direct WTP	5.30	numerator	n/a	graph anchor, derived

Component, millions of dollars	Amount	Benchmark Virginia/local share	Benchmark federal/outside share	Evidence and principal caveat
Indirect WTP	57.80	numerator	n/a	graph anchor, derived
CommonHelp development	10.779739	50%	50%	spending reported as of launch and treated by the paper as development cost; financing illustrative
Medicaid/FAMIS benefits	65.60	47.5%	52.5%	amount graph-derived; program mix open
Medicaid/FAMIS administration	-3.60	25%	75%	graph-derived; qualifying match open
SNAP benefits	58.50	0%	100%	amount graph-derived; federal financing verified
SNAP administration	-42.00	50%	50%	graph-derived and extrapolated from SNAP spending
TANF direct-cost residual	-3.979739	55%	45%	balancing residual; marginal block-grant incidence open
Medicaid/FAMIS externality	-37.10	0% in benchmark	100% balance	transported long-run saving; jurisdiction open
SNAP externality	1.00	included in 20% rule	balance	graph-derived

TANF/rounding externality	.485039	included in 20% rule		balance	calibrated residual

The direct-cost rows sum to \$85.30 million, and the externalities sum to -\$35.614961 million. The 47.5 percent medical-benefit share assumes that 15 percent of marginal medical enrollment is FAMIS. Regular Medicaid carried a 50 percent state share, while a time-weighted FAMIS calculation gives 33.23 percent over the evaluation window. The marginal FAMIS fraction is unreported. The 25 percent medical-administration share applies the enhanced federal rate for qualifying eligibility-system operation, whose eligibility has not been established for every affected dollar.

SNAP benefits were federally financed, with administration generally shared. TANF used a fixed federal block grant and state maintenance-of-effort spending. I use a 55 percent state share to approximate Virginia's reported basic-assistance accounting mix, though the opportunity cost of a marginal TANF dollar remains unresolved. CommonHelp also operated through local departments of social services. I combine Commonwealth and local accounts because I could not locate a local allocation for the causal period.

The benchmark direct Virginia/local cost is

$$\begin{aligned} G^{VA}_{direct} &= .50(10.779739) + .475(65.60) + .25(-3.60) \\ &\quad + 0(58.50) + .50(-42.00) + .55(-3.979739) \\ &= 12.461013. \end{aligned}$$

The remaining direct federal/outside cost is \$72.838987 million. The nonmedical externality is \$1.485039 million; assigning 20 percent to Virginia adds \$0.297008 million of Virginia cost. Total Virginia/local net cost is then \$12.758021 million, leaving federal/outside net cost of \$36.927018 million.

D.3 Resident scenarios and the medical-saving crossing

Scenario	Virginia/local net cost	Federal/outside net cost	Resident cost	State-resident result	Status
Population $\lambda =$ .0261299, no medical saving to Virginia	12.758	36.927	13.723	**4.60**	Derived, illustrative benchmark
Income-tax $\lambda =$ .0288382, otherwise same	12.758	36.927	13.823	**4.56**	Derived, illustrative

Scenario	Virginia/local net cost	Federal/outside net cost	Resident cost	State-resident result	Status
Direct-budget case, all behavioral externalities excluded	12.461	72.839	14.364	**4.39**	Different component ledger
Treasury limit	12.758	excluded	12.758	**4.95**	Budget-boundary limit
Consolidated check	n/a	n/a	49.685	**1.27**	Calibrated to author result
25% of medical saving assigned to Virginia	3.483	46.202	4.690	**13.45**	Illustrative
50% of medical saving assigned to Virginia	-5.792	55.477	-4.342	**pays for itself**	Illustrative

The population benchmark divides Virginia's population of 8,260,405 by the U.S. population of 316,128,839. The income-tax benchmark divides Virginia's $35.434 billion by the U.S. $1,228.719 billion in 2013 federal individual income tax after credits.

The main jurisdictional uncertainty is the modeled $37.1 million medical saving. Let s be the fraction assigned to Virginia, with the balance in the combined federal/outside account. Under the population weight,

$$D_{VA}(s) = 13.722919 - 37.1(1 - .0261299)s.$$

The denominator reaches zero at $s = 0.3798$, or about 38.0 percent. The income-tax weight gives a 38.4 percent crossing. The source assigns a dollar of long-run government medical saving to each dollar of childhood Medicaid/FAMIS spending by transporting estimates from another setting and extrapolating utilization through age 65. It leaves unidentified the child's later residence, the public payer that saves, the future match rate, and whether FAMIS entrants have the same effect. The crossing is a jurisdictional sensitivity and carries no confidence-interval or forecast interpretation.

A second sensitivity substitutes the 18.7 percent general-fund share reported later for a broader

Virginia eligibility-system portfolio for the benchmark 50 percent portal-development share. It also treats the TANF residual as fully controlled by Virginia. This gives Virginia/local net cost of about \$7.593 million, population-weighted resident cost of \$8.693 million, a ratio of about 7.26, and a medical-saving crossing near 24.1 percent. The evidence informs possible financing, though its broader portfolio is too distant from the 2012 CommonHelp portal to replace the benchmark.

Finally, the administrative-savings bars are modeled estimates without direct evidence of cash savings. Cholli and Wu estimate a proportional change in SNAP administrative cost per recipient and extend it to other programs. The \$42 million SNAP figure combines a graph reading with annual state-activity totals. These scenarios show which assumptions govern the result, while the available evidence supports no unique Virginia point estimate.

Appendix E. Additional participation cases

E.1 Massachusetts health-insurance automatic enrollment

Shepard and Wagner (2025) study Commonwealth Care applicants who had completed an application, been found eligible, and failed to choose a plan. Automatic assignment increased new enrollment by 48.4 percent relative to active enrollment alone. Suspending assignment reduced it by 32.6 percent relative to the automatic-enrollment baseline. The experiment therefore evaluates a post-eligibility default for people who had already applied.

The conventional welfare calculation uses one marginal passive-enrollee month as its unit. Insured medical claims are $C = 228.32$, modeled uncompensated care without insurance is $U = 135.313$, and enrollee WTP is $W = 93$. Shepard and Wagner assign 63.5 percent of uncompensated care to government and 36.5 percent to providers. Garthwaite, Gross, and Notowidigdo find that hospitals absorb approximately 60–67 percent, and 63.5 percent is their midpoint. Applying that evidence gives

$$G = .365U = 49.389, \qquad P = .635U = 85.924.$$

The consolidated numerator is $W + P = 178.924$, while the denominator is $C - G = 178.931$, giving approximately one. Because $G + P = U$, reversing the split leaves this ratio almost unchanged while reallocating WTP and fiscal savings. That reallocation changes the state-resident calculation, where government savings and provider welfare enter differently.

Let m_C be the marginal federal share of coverage claims, and let s and f be the portions of the government uncompensated-care offset realized as state and federal cash savings. Let χ be the share of provider/private welfare accruing to Massachusetts residents, and let A_S, A_F, A_L be omitted implementation cost. Then

$$\begin{aligned} D_T &= (1 - m_C)C - sG + A_S + A_L, \\ D_R &= D_T + \lambda(m_C C - fG + A_F), \\ N_R &= W + \chi P. \end{aligned}$$

The enrolled months span gross federal shares of 50, 58.78, 60.19, and 61.59 percent. Early months fell under a capped waiver pool, while later eligibility could be treated under Title XIX. The paper reports neither the month-by-category weights nor whether a marginal federal claim displaced another waiver expenditure. The HSN uncompensated-care account mixed hospital assessments, private-payer surcharges, balances, waiver-related funding, possible state appropriations, and provider shortfalls. Thus, the coverage match cannot identify its marginal cash allocation.

For comparison with the main text, suppose the full government uncompensated-care offset is realized at the same federal/state shares as coverage. I set $A_S = A_F = A_L = 0$ and use Massachusetts's 2009 income-tax share, $\lambda = .032281$. This gives

Gross coverage match m_C	Treasury denominator	Resident denominator	Resident ratio, $\chi = 0$	$\chi = .5$	$\chi = 1$
0	178.931	178.931	0.520	0.760	1.000
.5000	89.465	92.353	1.007	1.472	1.937
.5878	73.755	77.150	1.205	1.762	2.319
.6019	71.232	74.709	1.245	1.820	2.395
.6159	68.727	72.285	1.287	1.881	2.475

The $m_C = 0$ row is a capped-pool cash-incidence stress test. If an added CommCare claim displaces another valuable service, complete welfare accounting must also value that service.

Using the population share, $\lambda = .021246$, gives 1.018/1.488/1.958 at a 50 percent match and 1.309/1.913/2.518 at a 61.59 percent match. These are conditional scenarios because the marginal match remains unidentified.

The HSN allocation also matters. Holding fixed the income-tax weight and source-consistent split gives

Match	Government HSN offset treated as	Resident denominator	Resident ratios for $\chi = 0, .5, 1$
50.00%	no cash offset	117.845	0.789 / 1.154 / 1.518
50.00%	same match as coverage	92.353	1.007 / 1.472 / 1.937
50.00%	all state/local saving	68.456	1.359 / 1.986 / 2.614
50.00%	all federal saving	116.251	0.800 / 1.170 / 1.539
61.59%	no cash offset	92.237	1.008 / 1.474 / 1.940
61.59%	same match as coverage	72.285	1.287 / 1.881 / 2.475
61.59%	all state/local saving	42.848	2.170 / 3.173 / 4.176
61.59%	all federal saving	90.643	1.026 / 1.500 / 1.974

The reviewed sources do not identify the marginal-offset row. The paper also excludes the state-only noncitizen group from its main sample, leaving no basis for a fully state-funded calculation.

At the printed WTP, the source-consistent consolidated result is already 0.99996, so any positive omitted implementation cost moves it farther below one. Adding 4 or 4.5 percent of claims as a scale proxy gives 0.951 and 0.946. These proxies come from aggregate Connector administrative fees; the marginal cost of automatic assignment is unmeasured. I therefore report a lattice over the coverage match, HSN allocation, and provider incidence. The evidence does not select a point from it.

E.2 Virginia's ABAWD time limit

Gray et al. (2023) study Virginia's October 2013 reinstatement of the historical three-month-in-36-month SNAP time limit for able-bodied adults without dependents. Near the age-50 exemption, the rule reduced 18-month retention among incumbent participants by 23.4 percentage points from a no-rule retention rate of 63.2 percent. Eliminating it changes legal eligibility, work or activity conditions, documentation, and recertification. The counterfactual therefore extends beyond application simplification.

The welfare calculation evaluates elimination of the rule in 2018Q1 dollars per relevant ABAWD-month. Its central ledger is

Component	WTP	Consolidated cost	Federal	Commonwealth	Virginia local
Restored SNAP benefits	74.47	74.47	74.47	0	0
Avoided new applications	0	-4.26	-2.13	-1.4697	-.6603
Added recertifications	0	1.84	.92	.6348	.2852
Net administration	0	**-2.42**	**-1.21**	**-.8349**	**-.3751**
Authors' central earnings-related tax loss	0	11.03	11.03	0 benchmark	0
Additional bounded work-effort value	0 to 42.24	0	0	0	0

Component	WTP	Consolidated cost	Federal	Commonwealth	Virginia local
Central total	**74.47 to 116.71**	**83.08**	**84.29**	**-.8349**	**-.3751**

The restored-benefit term is (.234/.632)(201.12) = 74.47. Avoided applications save \$4.26, and additional recertifications cost \$1.84, for a net administrative saving of \$2.42. The \$11.03 tax term uses suggestive 24-month earnings results; the average at 18 months is statistically null. These inputs give a central consolidated range of 0.90–1.40. Setting the earnings-related tax effect to zero gives 1.03–1.40, while an adverse calibration gives 0.81–1.27.

SNAP benefits were federally financed. The historical administrative split was 50 percent federal, 34.5 percent Commonwealth, and 15.5 percent local. With no net Virginia income-tax response for the representative low-income filer, state and local government save \$1.21 per relevant month. The treasury account therefore **pays for itself on the measured components**.

Let λ be resident incidence of federal cost. With federal costs of \$73.26, \$84.29, and \$92.86 in the zero, central, and adverse earnings cases,

$$D_{VA} = -1.21 + \lambda G_F.$$

Earnings case	WTP	Resident denominator, population .0260812	Resident ratio	Denominator, tax share .0280391	Resident ratio
Zero	74.47–100.62	.7007	106.28–143.59	.8441	88.22–119.19
Central	74.47–116.71	.9884	75.35–118.08	1.1534	64.56–101.19
Adverse	74.47–116.71	1.2119	61.45–96.30	1.3937	53.43–83.74

The central denominator changes sign at $\lambda = .01436$. The reported 64.56–118.08 range combines two central denominators with the WTP endpoints. Since all of these ratios divide by roughly one dollar, the denominators provide the more stable information. The certification costs are back-of-the-envelope averages. Systems programming, notices, guidance, training, exemption screening, appeals, and work-program capacity are omitted from the calculation. A transition cost of one dollar per normalized unit would roughly double the central resident denominator.

The live Policy Impacts page reverses digits in \$74.47, describes retained incumbents as new participants, labels the 23.4-point retention effect as general participation, and reports 1.02 for the zero-earnings lower endpoint. The official online appendix gives 1.03, which is the value used here.

E.3 Mandatory SNAP Employment and Training for parents

Cook and East (2026) study parents already receiving SNAP in an unidentified mountain-plains state. A household head could be referred to mandatory Employment and Training when their youngest child turned six shortly before recertification. The IV estimate implies that an induced referral reduces household SNAP benefits by \$669 over six months. The welfare counterfactual eliminates mandatory E&T and thereby restores \$111.50 per month. It assigns the recipient \$111.50 in WTP, sets the earnings-related fiscal effect to zero, and subtracts \$11.32 in planned average E&T cost. This gives

$$M_N = \frac{111.50}{111.50 - 11.32} = 1.113.$$

The \$11.32 consists of \$9.42 for supervised job search and \$1.90 for participant reimbursements. Historical rules divide the reimbursement line 50/50 between federal and nonfederal funds. Supervised job search could draw on the finite 100 percent federal formula grant, 50/50 reimbursement, or a mixture of the two. The paper leaves this mix unreported. A state, locality, workforce entity, community college, or nonprofit could provide the nonfederal match.

Let s denote the share of job search paid from the 100 percent federal grant. Let q denote the share of that grant expenditure that becomes an actual federal cash saving when E&T is eliminated. Federal and nonfederal savings are then

$$S_F = .95 + 4.71(1 - s) + 9.42qs, \qquad S_N = .95 + 4.71(1 - s).$$

Next, let δ be the share of the nonfederal saving accruing to state residents. Using the historical zero direct state benefit share, the paper's WTP, and zero omitted fiscal and transition effects gives

$$D_R = \lambda(111.50 - S_F) - \delta S_N.$$

The core grid sets $q = 1$ and $\delta = 1$.

λ	All job search 50/50, $s = 0$	Half formula, $s = .5$	All formula, $s = 1$
0	pays for itself, $D = -5.660$	pays for itself, $D = -3.305$	pays for itself, $D = -.950$
.005	pays for itself, $D = -5.131$	pays for itself, $D = -2.788$	pays for itself, $D = -.444$
.010	pays for itself, $D = -4.602$	pays for itself, $D = -2.270$	1,818.92, $D = .061$
.020	pays for itself, $D = -3.543$	pays for itself, $D = -1.235$	**103.95,** $D = 1.073$
.050	pays for itself, $D = -.368$	59.65, $D = 1.869$	27.15, $D = 4.106$
.100	22.64, $D = 4.924$	15.83, $D = 7.044$	12.17, $D = 9.163$
1	1.11, $D = 100.180$	1.11, $D = 100.180$	1.11, $D = 100.180$

As s moves from zero to one, the zero crossings are $\lambda = .05348$, $.03194$, and $.00939$. At $\lambda = .02$, changing δ from zero to one moves the all-formula result from 55.13 to 103.95. If formula funding is fully reallocated, the cash-ledger denominator rises from 1.073 to 1.261 and the ratio falls from 103.95 to 88.42. A complete welfare calculation for this reallocation would also include the value of the grant's alternative use. All of the very large ratios arise from denominators close to zero.

The \$11.32 comes from a later state plan and measures an average. The marginal saving per induced referral remains unmeasured, and the plan omits some staffing. The calculation also assigns zero to the value of avoiding the E&T ordeal, the private value of any lost voluntary service, labor-market fiscal effects, and transition costs. Adding \$5 of resident-weighted transition cost to the 103.95 scenario lowers it to 18.36; adding \$20 lowers it to 5.29. A point estimate would require the state plan's funding columns and a consistent policy unit.

E.4 SNAP caseworker effectiveness

A separate paper by Cook and East (2025) uses quasi-random assignment of applicants in an anonymous mountain-plains state to SNAP caseworkers with different conditional approval rates. A one-standard-deviation increase in the caseworker index raises receipt in the application quarter by 2.68 percentage points and reduces incomplete applications by 1.27 points. These estimates are consistent with caseworkers helping complete applications that have already been started. The study leaves open which intervention, including training, staffing, software, or monitoring, would produce the one-standard-deviation shift.

For a hypothetical one-standard-deviation improvement, the unrounded three-year inputs per exposed applicant are

Component	Formula	Amount
Added benefits and paper WTP, B	683(.03)	20.4900
Added recipient-quarters	2.657(.03)	.07971
Routine administration, A	(261/4)(2.657)(.03)	5.2011
Earnings response, Y	1,596(.03)	47.8800
Paper fiscal saving	$.40Y$	19.1520
Net government cost	$B + A - .40Y$	6.5391
Reconstructed ratio	20.49/6.5391	**3.13**

The paper rounds the numerator to \$20, direct cost to \$25, and fiscal saving to \$19, then reports $20/(25-19) = 3.3$. Thus, 3.3 is the author-reported value and 3.13 is the unrounded reconstruction. The 40 percent fiscal rate combines a 24 percent SNAP phase-out, employee payroll tax, and a residual income-tax calculation with inconsistent filing conventions. The benefit outcome may already incorporate the SNAP phase-out, making the separate 24 percent offset a possible double count.

Historically, benefits were federal and ordinary administration was approximately 50/50. Under the paper's taper convention, with zero implementation and state-tax effects, the accounts are

$$G_{SL} = 2.6005, \qquad G_F = 3.9385.$$

Let K be implementation resource cost per induced completed application. Since the incompletion effect is .01272 per exposed applicant, each \$100 of K adds \$1.272 before financing weights. Assuming 50/50 financing for implementation and routine administration gives the following grid of resident denominators and ratios.

λ	$K = 0$	$K = 100$	$K = 500$
0, treasury limit	2.60 / 7.88	3.24 / 6.33	5.78 / 3.54
.002	2.61 / 7.86	3.25 / 6.31	5.79 / 3.54
.005	2.62 / 7.82	**3.26 / 6.29**	5.82 / 3.52
.010	2.64 / 7.76	3.28 / 6.24	5.85 / 3.50
.020	2.68 / 7.65	3.33 / 6.16	5.92 / 3.46
1, consolidated	6.54 / 3.13	7.81 / 2.62	12.90 / 1.59

The 6.29 reference uses $\lambda = .005$, full transfer valuation, a \$100 implementation cost per induced completion, no state tax effect, and a 50/50 implementation match. This is an illustrative calibration because the state and production technology remain unidentified. Halving transfer WTP and using $\lambda = .02$, $K = 500$ produces 1.73. With the same implementation cost entirely state funded, the ratio is 1.13. An implementable state-resident MVPF requires evidence on both the state and the intervention.

The live Policy Impacts page combines two papers. Its introduction describes the caseworker study, while the E&T worksheet, journal citation, and DOI come from the parents study. Every checked public version of the parents paper uses \$669 and \$111.50; the worksheet uses \$708 and \$118. The caseworker paper's own welfare result is 3.3. The two policies must remain separate.

Appendix F. Source and version record

F.1 Inclusion and source rule

I include a study when it evaluates a U.S. public-transfer policy or participation mechanism, treats enrollment, retention, application completion, or continued participation as a primary outcome, and supplies either an MVPF or a sufficiently explicit welfare calculation. Cases with an incomplete policy boundary or state allocation remain in the additional-cases section, with the missing information stated directly.

The published article or dated working-paper version, its online appendix, and contemporaneous program-financing sources govern the calculations. I use Policy Impacts to discover cases and compare reported results, but resolve discrepancies using the underlying studies and authoritative program sources.

F.2 Included studies and versions

Study and policy or welfare counterfactual	Version used	Policy Impacts page	Important version or transcription note
Finkelstein and Notowidigdo (2019), information and assistance for likely eligible Pennsylvania older adults	*Quarterly Journal of Economics* 134(3), 1505–1556, and online appendix	Information; assistance	Information-page dollar fields are inconsistent; the assistance page omits the costless/resource-inclusive distinction
Unrath (2024), removing a quarterly CalFresh report	March 2024 manuscript	Removing SNAP Reporting Requirements	Figure 13 gives 4.25; printed page-23 inputs give 4.33; live library page retains older 2.09
Cholli and Wu (2026), Virginia CommonHelp	June 18, 2026 working paper distributed by the Washington Center for Equitable Growth in July 2026	No standalone page located as of August 18, 2026	Exact 1.27 and launch spending are reported, but aggregate WTP, direct-cost, and externality totals are graph-only and no numerical workbook was located
Shepard and Wagner (2025), Commonwealth Care automatic assignment	*American Economic Review* 115(3), 772–822, and online appendix	Automatic Enrollment in Health Insurance	Paper and library invert the source direction of hospital versus government uncompensated-care incidence
Gray et al. (2023), eliminating Virginia's ABAWD time limit	*American Economic Journal: Economic Policy* 15(1), 306–341, and online appendix	Work Requirements for SNAP	Library has WTP, classification, and metadata errors; the article's Appendix D.7 supplies the complete welfare ledger
Cook and East (2026), eliminating mandatory E&T for parents	May 27, 2025 author manuscript corresponding to *Journal of Public Economics* 257, article 105631	SNAP Caseworker Effectiveness, invalid combined record	Current public versions use $669/$111.50; the page uses $708/$118 and combines two papers

Study and policy or welfare counterfactual	Version used	Policy Impacts page	Important version or transcription note
Cook and East (2025), hypothetical increase in SNAP caseworker effectiveness	NBER Working Paper 31307, revised May 2025	Same invalid combined record	Paper reports 3.3 and unrounded inputs give 3.13; the library's E&T worksheet is a stale, unsupported splice from the parents calculation

F.3 Primary financing and incidence sources

I allocate government costs using the rules in effect during each evaluation. USDA Food and Nutrition Service program and administrative-cost sources provide the SNAP benefit and historical administrative shares. Contemporaneous Assembly Budget Committee and California Department of Social Services documents provide California's 50/35/15 administrative division. HHS and CMS tables give the Virginia Medicaid/FAMIS match rates. Administration for Children and Families financial reports govern the TANF block-grant and maintenance-of-effort treatment. For Massachusetts coverage and HSN, I use waiver documents, Connector annual reports, and the HSN annual report.

References

Cholli, N., and D. Wu. 2026. "Multiple Program Participation in the Safety Net: Incidence, Impediments, and Implications." Washington Center for Equitable Growth Working Paper.

Cook, J. B., and C. N. East. 2025. "The Effect of Means-Tested Transfers on Work: Evidence from Quasi-Randomly Assigned SNAP Caseworkers." NBER Working Paper 31307, revised May 2025.

Cook, J. B., and C. N. East. 2026. "The Disenrollment and Labor Market Effects of SNAP Work Requirements on Parents." *Journal of Public Economics* 257: 105631.

Finkelstein, A., and M. J. Notowidigdo. 2019. "Take-Up and Targeting: Experimental Evidence from SNAP." *Quarterly Journal of Economics* 134(3): 1505–1556.

Garthwaite, C., T. Gross, and M. J. Notowidigdo. 2018. "Hospitals as Insurers of Last Resort." *American Economic Journal: Applied Economics* 10(1): 1–39.

Gray, C., A. Leive, E. Prager, K. Pukelis, and M. Zaki. 2023. "Employed in a SNAP? The Impact of Work Requirements on Program Participation and Labor Supply." *American Economic Journal: Economic Policy* 15(1): 306–341.

Hendren, N., and B. Sprung-Keyser. 2020. "A Unified Welfare Analysis of Government Policies." *Quarterly Journal of Economics* 135(3): 1209–1318.

Shepard, M., and M. Wagner. 2025. "Do Ordeals Work for Selection Markets? Evidence from Health Insurance Auto-Enrollment." *American Economic Review* 115(3): 772–822.

Unrath, M. 2024. “Targeting, Screening, and Retention: Evidence from the Supplemental Nutrition Assistance Program in California.” Working paper, March 2024.